FATMA AHMED HEGAZY

AI & Machine Learning Student

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PROFESSIONAL SUMMARY

Fourth-year Artificial Intelligence student at Menoufia University, specializing in Machine Learning and Data Analysis. Proficient in Python, Power BI, and deep learning frameworks with hands-on experience building predictive models, NLP pipelines, unsupervised learning systems, reinforcement learning agents, and interactive dashboards. Recognized for analytical thinking, cross-functional collaboration, and a proactive approach to problem-solving.

EDUCATION

Bachelor of Artificial Intelligence — Machine Learning Track 2023 – Present

Faculty of Artificial Intelligence, Menoufia University

- Term GPA 3.64 / 4.0
- Cumulative GPA 3.27 / 4.0

TECHNICAL SKILLS

Programming: Python (Advanced) | Java (Intermediate) | C++ (Intermediate) | SQL (Basic)
Data Analysis: Pandas | NumPy | Scikit-learn | Matplotlib | Seaborn
Visualization: Power BI | DAX | Data Modeling | Tableau (Basic)
ML / DL: Regression | SVM | Logistic Regression | LightGBM | CNN | PyTorch | K-Means | DBSCAN
NLP: TF-IDF | Sentiment Analysis | ABSA | NLTK | spaCy | Text Preprocessing
Reinforcement Learning: Q-Learning | SARSA | Custom Grid Environments | Reward Shaping
Computer Vision: OpenCV | Pillow
Dev Tools: Jupyter Notebook | Google Colab | VS Code | Streamlit | Tkinter | Maven | JUnit 5

PROJECTS

Credit Card Customer Segmentation | Python | K-Means | DBSCAN | GMM | PCA | Streamlit *2025–2026*

- Designed a robust unsupervised learning pipeline on 9,000 credit card records, comparing K-Means, DBSCAN, GMM, and Agglomerative Clustering.
- Applied Winsorization, robust scaling, skewness correction, and PCA for dimensionality reduction; improved Silhouette Score from 0.14 to 0.31.
- Engineered business-facing cluster profiles with actionable customer segment labels and delivered a Streamlit app for real-time inference.

Loan Default Prediction System | Python | LightGBM | SMOTE-ENN | Optuna | Streamlit *2025–2026*

- Built an end-to-end credit risk pipeline on the Give Me Some Credit Kaggle dataset (150K rows); addressed severe class imbalance with SMOTE-ENN.
- Tuned LightGBM with Optuna (100 trials), built a stacking ensemble, and applied cost-sensitive threshold optimization to maximize business utility.
- Deployed a dual-mode Streamlit GUI supporting single-customer and batch CSV prediction with downloadable risk reports.

Aspect-Based Sentiment Analysis (ABSA) | Python | TF-IDF | Logistic Regression | NLTK | Streamlit *2025–2026*

- Developed a binary sentiment classifier for customer reviews using TF-IDF and sentiment lexicon feature engineering with Logistic Regression.
- Applied stratified cross-validation, class weight balancing, and custom text preprocessing pipeline (tokenization, stopword removal, lemmatization).
- Built a Streamlit analytics platform with real-time aspect extraction, interactive sentiment visualizations, and exportable reports.

Reinforcement Learning — Vacuum Cleaner Agent | Python | Q-Learning | SARSA | Custom Environment | GIF Animation *2025–2026*

- Implemented and compared Q-Learning (off-policy) vs SARSA (on-policy) on a custom 8x8 grid vacuum cleaning environment built from scratch.
- Applied reward shaping, epsilon-greedy exploration, and fixed state encoding; visualized learning curves, policy maps, and GIF trajectory animations.
- Analyzed convergence behavior and performance differences between off-policy and on-policy temporal difference methods.

Car Dealership Management System | Java 17 | OOP | SOLID | Design Patterns | JUnit 5 | Maven *2025–2026*

- Built a Java OOP application applying all four OOP pillars and all five SOLID principles through a modular car dealership domain model.
- Implemented three design patterns: Factory Pattern (object creation), Strategy Pattern (swappable discounts), and State Pattern (car lifecycle).
- Wrote 35+ JUnit 5 unit tests covering constructors, getters/setters with validation, discount strategies, factory creation, state transitions, and edge cases.

Sales & Business Intelligence Dashboard | Power BI | DAX | Data Modeling *2024–2026*

- Built a multi-page Power BI report analyzing sales, customers, products, and territories using the AdventureWorks 2022 dataset.
- Designed relational star-schema data models and created advanced DAX measures for KPIs, YoY comparisons, and trend analysis.
- Delivered interactive visual storytelling enabling data-driven decision-making across key business functions.

Breast Cancer Detection using Deep Learning | PyTorch | CNN | OpenCV *2024*

- Developed a CNN-based image classification model achieving 85–90% accuracy on medical imaging data.
- Applied image preprocessing, data augmentation, and hyperparameter tuning; evaluated with confusion matrix and classification reports.

Medical Diagnosis Prediction | Scikit-learn | SVM | Logistic Regression *2024*

- Built ML models for disease prediction, improving accuracy from 78% to 89% through feature engineering and hyperparameter tuning.

CERTIFICATIONS

Introduction to Artificial Intelligence	*NTI* | *1 Month*
Artificial Intelligence for Business	*ITIDA / NTI* | *1 Month*
Machine Learning	*NTI* | *1 Month*
Computer Vision	*NTI* | *1 Month*

VOLUNTEER EXPERIENCE & LEADERSHIP

HR Committee Member — YLY (Youth Leading Youth) 2024–2025

Ministry of Youth & Sports, Menoufia

- Completed the Youth Leadership Program (YLP) training; contributed to HR operations within a government-backed youth development initiative.

HR Member — Computer Family 2023–Present

Faculty of Artificial Intelligence, Menoufia University

- Supported team coordination, event logistics, and student engagement activities within the faculty's student technology community.

ADDITIONAL INFORMATION

Languages: Arabic (Native) | English (Upper-Intermediate)
Soft Skills: Analytical Thinking | Team Collaboration | Fast Learner | Cross-functional Collaboration
Work Style: Remote-ready | Cairo-available
Location: Menoufia, Egypt